Hangzhou Hikrobot Technology Co.,Ltd.

# Hikrobot Firmware Upgrade Tool

## User Manual

HIKROBOT

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# Chapter 1 Legal Information

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## About this Manual

The Manual includes instructions for using and managing the Product. Pictures, charts, images and all other information hereinafter are for description and explanation only. The information contained in the Manual is subject to change, without notice, due to firmware updates or other reasons. Please find the latest version of this Manual at the Hikrobot website (***https://en.hikrobotics.com***).
Please use this Manual with the guidance and assistance of professionals trained in supporting the Product.

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# Chapter 2 Disclaimer

The manual guides you to establish and configure the Software. To ensure the properness of usage and stability of the Software, refer to the contents below and read the manual carefully before installation and operation.
As GIFs and videos cannot be played in PDF document, you are recommended to read the online document.

## 2.1 Get Support

You can get support from us in the following ways:
- **Official Website**: visit ***https://en.hikrobotics.com/*** to get other related documents.
- **Email**: global.support@hikrobotics.com.

## 2.2 Symbol Conventions

The symbols that might be found in this document are defined as follows.

| Symbol | Description |
|---|---|
| **Note** | Provides additional information to emphasize or supplement important points of the main text. |
| **Caution** | Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance degradation, or unexpected results. |
| **Danger** | Indicates a hazardous situation which, if not avoided, will or could result in death or serious injury. |

# Chapter 3 Release Notes

Key updates of each version are as follows.

## V4.4.0

This version was released on August, 2024.

- Changed **Camera Link** interface to **Serial Port** and **GenTL** to **PCIe**.
- Added **Fuzzy Matching** to frame grabber upgrade page. See ***Upgrade a Frame Grabber***.

## V4.3.0

This version was released on October, 2023.
Optimized the operations of upgrading and configuring frame grabbers. See ***Upgrade a Frame Grabber***.

## V4.2.0

This version was released on June, 2023
You can use the Firmware Upgrade Tool to configure firmware upgrading settings and upgrade the firmware of cameras and frame grabbers.

# Chapter 4 Tool Introduction

You can use the Firmware Upgrade Tool (hereinafter simplified as "the Tool") to upgrade the firmware of GigE cameras, USB cameras, PCIe cameras, CoaXPress cameras, and frame grabbers.
The main window is as below and the descriptions are in the table.

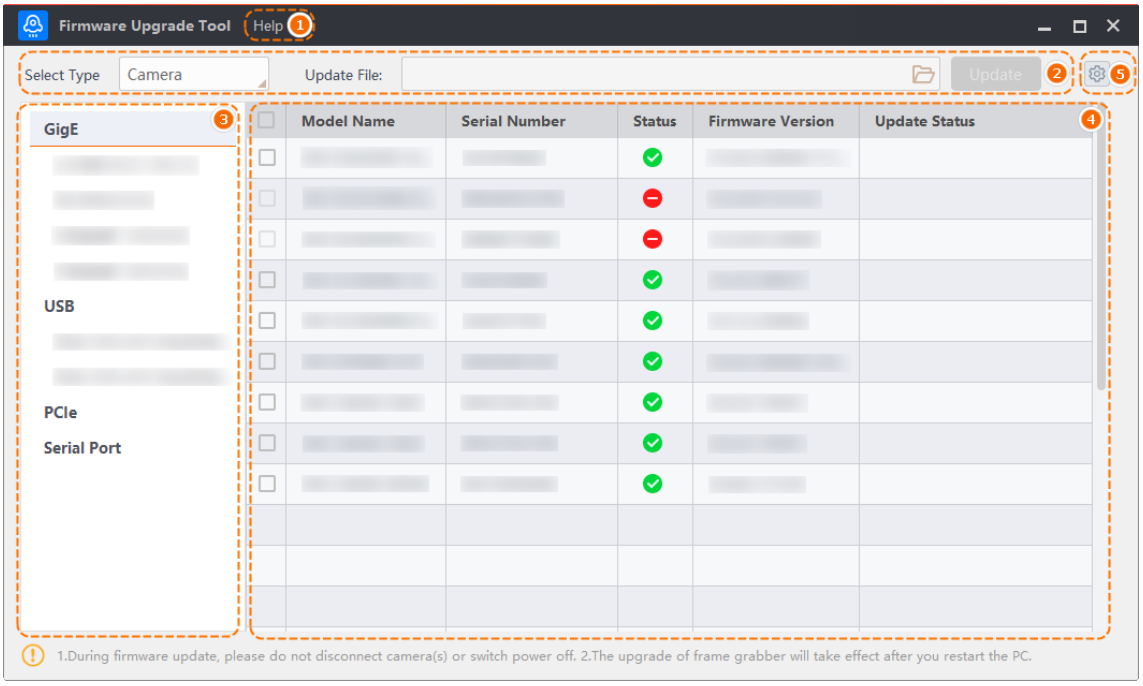


**Figure 4-1 Firmware Upgrade Tool**

**Table 4-1 Main Window Description**

| Number | Area | Description |
|---|---|---|
| 1 | Help | You can select a language (English or Chinese), and view the version information. |
| 2 | Upgrading Configuration | • You can select camera or frame grabber as the to-be-upgraded device type.<br>• You can select a camera upgrade package in area 2, and you can select a frame grabber upgrade package in area 4. |
| 3 | Interface Information | You can view the interface information of your computer. The available operations may vary according to devices with different interfaces. See ***Upgrade a Camera***. |

| Number | Area | Description |
|---|---|---|
| 4 | Device Information | You can view devices and related information of different interfaces. |
| 5 | Device List Configuration | You can set what kind of information will be displayed in area 3. The available information may vary according to interfaces. |

# Chapter 5 Upgrade Process

You can upgrade the firmware of cameras and frame grabbers.

## 5.1 Upgrade a Camera

The camera firmware upgrading process mainly contains three parts, and they are Search for Device, Start to Upgrade, and View Upgrading Status.

### Search for Devices

After you open the Tool, you can select a camera in the **Select Type** area, and you can see the computer interface information displayed on the left. The available operations may vary according to devices with different interfaces.

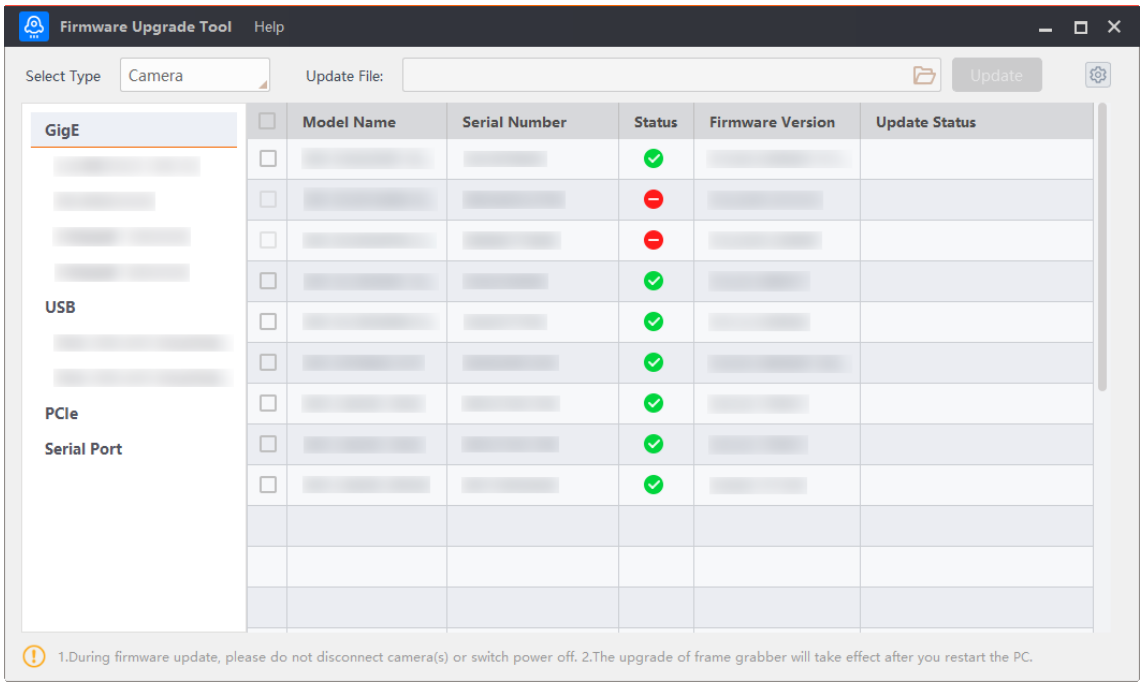


**Figure 5-1 Upgrade a Camera**

### Upgrade Camera Firmware

- GigE and USB Interfaces:
  - Select **GigE** or **USB**, and the devices that can be found will be displayed on the right pane.
  - Select an interface under GigE or USB, and the devices that can be found will be displayed on the right pane.
  - The Tool can automatically refresh the devices under the enumeration of GigE and USB, or click ↻ to manually refresh the enumeration.

- PCIe Interface:
  - Select **PCIe**, and the devices that can be found will be displayed on the right pane.
  - Select an interface under PCIe, and the devices that can be found will be displayed on the right pane.
  - The Tool can automatically refresh the devices under the enumeration of PCIe, or click to manually refresh the enumeration.
- Serial Port Interface:
  - Select **Serial Port**, and the available devices will be displayed on the right pane.
  - Select an interface from Serial Port, and the available devices of the interface will be displayed on the right pane.
  - By default, the camera information under Serial Port will not be refreshed automatically. You can click on the right side of the Serial Port to refresh the enumeration manually.

## Start to Upgrade

Check if the camera to be upgraded is available. Click to select a firmware upgrade package (dav file) in the upper right side of the Tool.
The Tool can upgrade the firmware of multiple cameras in a batch. Up to 20 cameras can be selected at the same time.

- If the upgrade package is for a specific model, only cameras of the same model can be upgraded in batch. For cameras of other models, the status bar will prompt "Upgrading failed. (Error code: 0x900006500) Firmware mismatch.".
- If the upgrade package is for multiple models, you can upgrade the cameras of multiple models in the upgrade package. For cameras of the other models that are not included in the upgrade package, if you upgrade them, the status bar will prompt "Upgrading Failed. (Error code: 0x900006500) Firmware mismatch.".

After selecting the firmware upgrading package, click **Upgrade**.

**Note**

- Do not disconnect the camera from the PC during firmware upgrade.
- The camera will reboot automatically after upgrading.

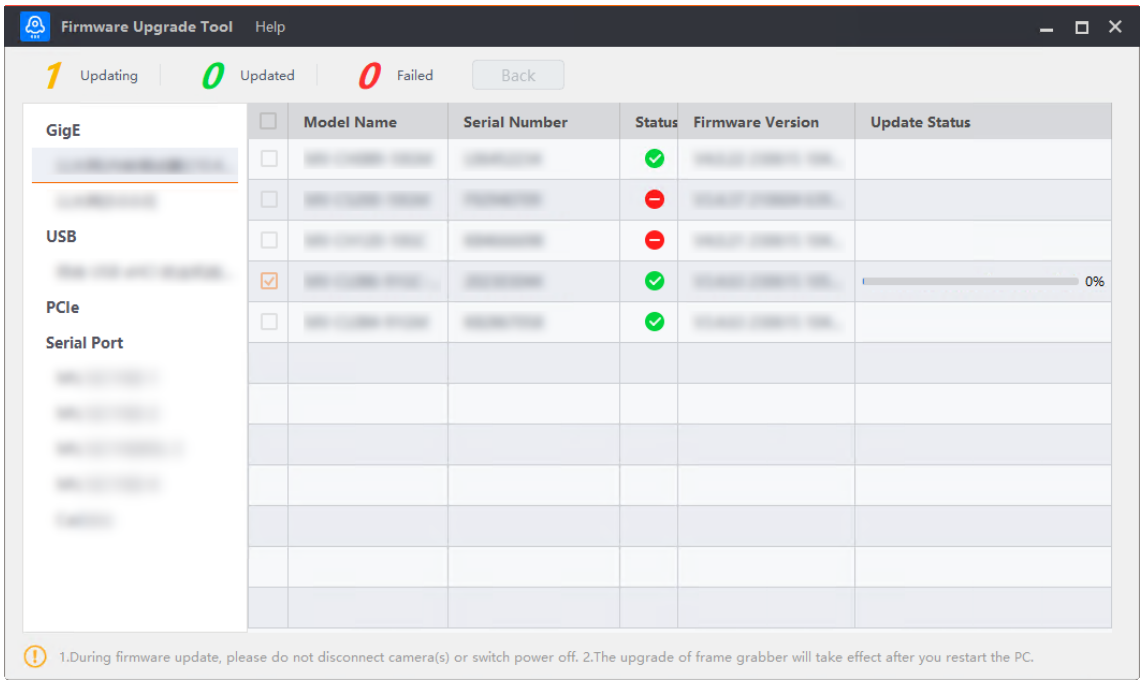


**Figure 5-2 Upgrading Process**

### View Upgrading Status

In the upper-left corner, the upgrading information is displayed as shown below. You can click **Back** on the top of the Tool to go back to the initial interface. Also, on the right of the Tool, the upgrading status information of a selected device is displayed.

## 5.2 Upgrade a Frame Grabber

The frame grabber firmware upgrading process mainly contains three parts, and they are Search for Device, Start to Upgrade, and View Upgrading Status.

### Search for Devices

After you open the tool, select **FrameGrabber** in **Select Type**, and all connected PCIe devices displays on the left.

**Note**

By default, the Tool automatically enumerates connected frame grabbers every 30 seconds. You can also click ↻ to re-enumerate.

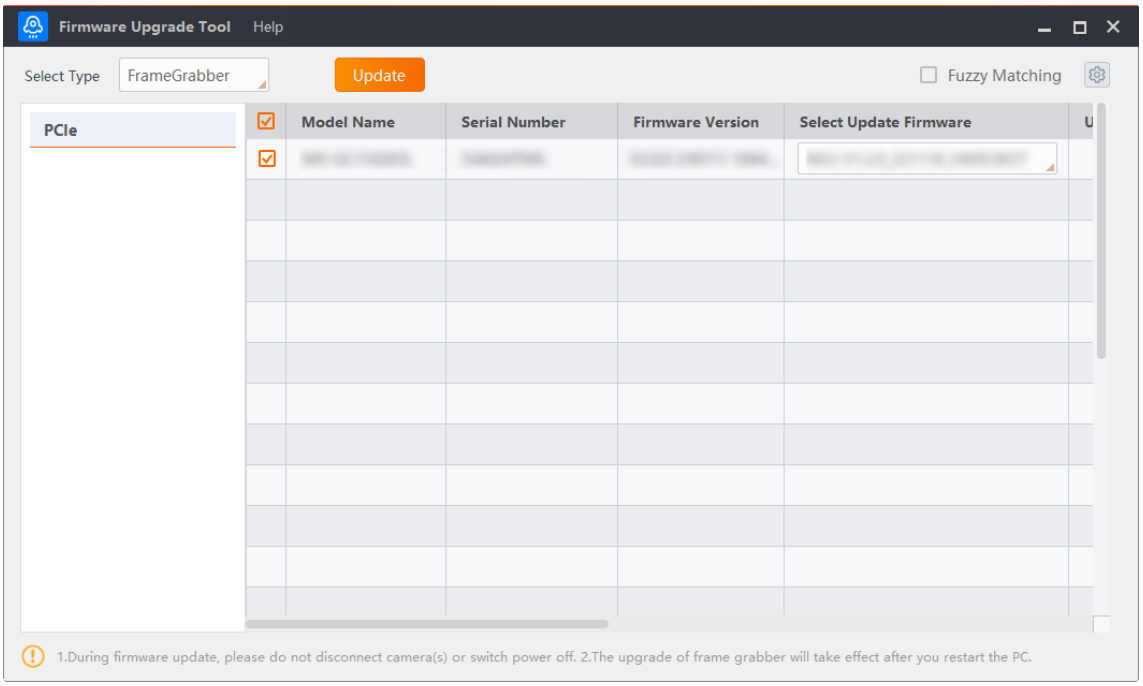


**Figure 5-3 Upgrade Frame Grabber Firmware**

## Start to Upgrade

Please visit Hikrobot official website to get the firmware driver packages of the to-be-upgraded frame grabbers.
Before upgrading, ensure the frame grabber(s) to be upgraded are available during upgrading process.

1. Install the firmware driver under the directory: C:\Program Files (x86)\Common Files\MVS\FirmWare.
   Once installed, the Tool detects and displays available firmwares, as shown below.

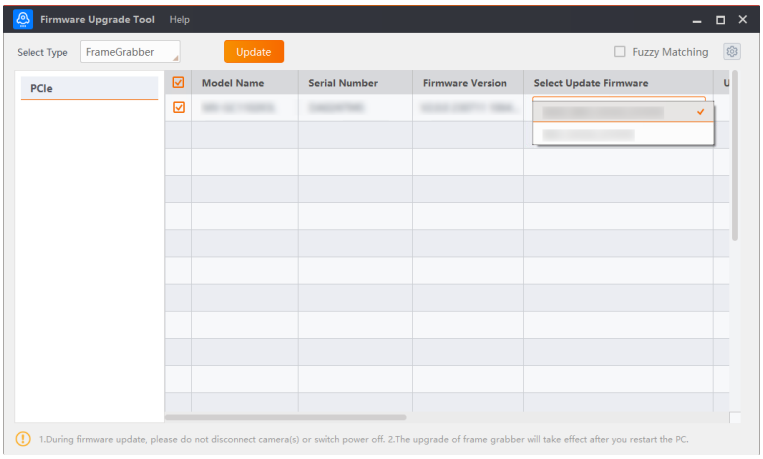


**Figure 5-4 Upgrade Frame Grabber**

2. Select frame grabber(s) to be upgraded.
3. Select required firmware in **Select Update Firmware** for each frame grabber.

**Note**

- The Tool supports upgrading up-to-20 frame grabbers in a batch.
- The Tool supports fuzzy searching of firmware files. To enable this functionality, enable **Fuzzy Matching**.

4. Click **Upgrade** to upgrade them.

**Note**

- Keep the device(s) connected when upgrading.
- The frame grabber(s) will reboot automatically when upgraded.

## View Upgrading Status

Once the upgrading starts, you can view the status on top, as shown below. The upgrading progress also displays in **Update Status**.
In case of upgrading failure, you can click **Back** to return to the initial window and redo the upgrading.

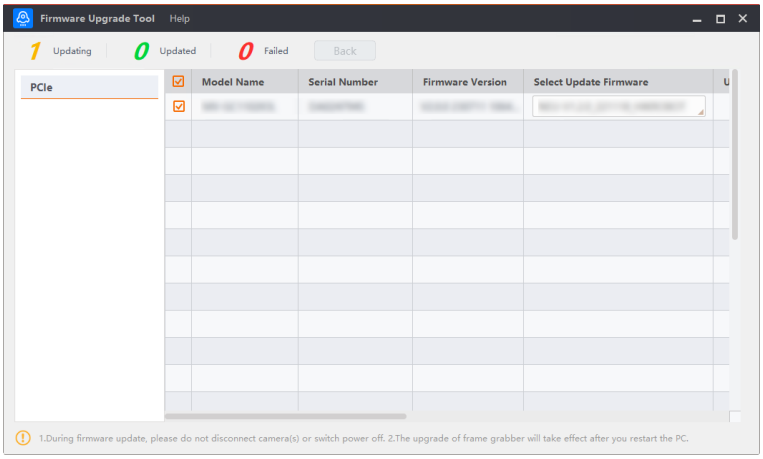


**Figure 5-5 Upgrading Frame Grabber**

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